eBay Case Study: From Salvage Yard to Global Marketplace

Build an AI assistant that helps an automotive salvage seller prepare accurate used-parts listings for international buyers.

1. Business context

This challenge focuses on an offline seller in the **Parts & Accessories (P&A) category**, dealing in used automotive parts recovered from dismantled vehicles.

The seller's inventory information is fragmented across photographs, spreadsheets, handwritten labels, and donor-vehicle records. Preparing an international listing requires bringing this information together: identifying the part, documenting its condition, confirming available identifiers, and establishing what compatibility information can be supported.

The challenge is not simply to generate a product description. It is to help the seller distinguish **what is known, what needs confirmation, and what should not be claimed without further evidence**.

The seller and all examples below are fictional scenarios designed for the hackathon.

2. Meet the seller

Noor runs a Dubai-based automotive salvage business with an initial inventory of 120 used parts, including headlights, taillights, side-view mirrors, door handles, and interior trim recovered from dismantled vehicles.

She sells primarily to local garages and traders through yard visits, phone calls, and WhatsApp enquiries. Her team understands the inventory but has limited experience creating listings for international buyers.

Product photographs are stored on employees' phones, while a basic spreadsheet records short descriptions, asking prices, and storage locations. Donor-vehicle details are captured inconsistently, part numbers are sometimes missing or difficult to read, and testing status and cosmetic damage are not always documented. Some additional information is available only in Arabic notes.

Noor wants to prepare her first ten parts for sale on eBay. She needs an assistant that can organise the available information, ask targeted questions, and produce listing drafts without guessing vehicle compatibility or overlooking condition issues.

3. The challenge

How might AI help an offline automotive salvage seller turn fragmented inventory information into accurate, compatibility-aware used-parts listing drafts and a clear cross-border readiness checklist without inventing part details, fitment, or condition claims?

What participants should build

Develop a working prototype that supports the following journey:

Stage	Expected capability
Understand the inventory	Accept photographs, spreadsheets, and donor-vehicle notes. Create a structured record for each part, capturing supported information such as part type, manufacturer, part number, donor vehicle, condition, and storage location. Retain the source of each attribute.
Resolve information gaps	Detect missing or conflicting details. Ask focused questions about identifiers, left/right position, visible damage, testing status, included components, and packaged dimensions and weight. Support English and one additional language, preferably Arabic.
Prepare listing drafts	Generate an editable title, description, category suggestion, and relevant item specifics. Clearly disclose used condition, defects, and testing status. Separate donor-vehicle information from independently supported compatibility information.
Guide the next steps	Check each part against a supplied listing and shipping-readiness checklist. Classify it as **ready for seller review**, **information missing**, or **specialist review required**, with a prioritised list of actions.

A strong solution should let Noor correct information once and carry that correction through the inventory record, listing draft, and readiness checklist.

Illustrative demonstration

Noor uploads photographs of a used side-view mirror. The spreadsheet describes it as **“Toyota Corolla mirror, 2017,”** while the dismantling note records a **2018 donor vehicle**. The part-number label is partially unreadable, one photograph appears to show a damaged mounting tab, and electrical testing has not been recorded.

The assistant should flag the conflicting donor-vehicle details, request a clearer photograph of the identifier, and ask Noor to confirm the mounting-tab condition and testing status.

It should **not** infer a broad range of compatible vehicles from appearance alone, describe the mirror as fully functional without evidence, or omit the possible damage to make the listing more attractive.

After Noor confirms the details, the assistant should update the draft and explain which compatibility claims remain unverified.

4. Three-day prototype scope

Limit the core demonstration to **ten used parts within one product family**, such as side-view mirrors, and **one test trade corridor**, such as UAE-to-UK. Use a clearly labelled test checklist rather than attempting comprehensive customs or regulatory assessment.

Assume seller registration is complete. Live publication, actual shipment booking, and production-system integration are outside the core scope. Airbags, batteries, and other hazardous or safety-critical components should be excluded from this prototype challenge.

Suggested input pack: Twenty fictional part records, accompanying photographs, multilingual inventory notes, donor-vehicle records, a small approved compatibility reference dataset, and a versioned listing/shipping checklist. Include missing identifiers, conflicting vehicle details, visible defects, and untested components, with a separate answer key for judges.

5. How success should be demonstrated

Teams should compare their prototype with a manual workflow using the same sample inventory. Measure preparation time, accuracy of populated attributes, detection of missing or conflicting information, completeness of condition disclosures, and corrections required before seller approval.

The intended business value is faster seller activation and more accurate, discoverable used-parts inventory not simply more generated listings. A draft that correctly flags uncertain compatibility should score better than one that appears complete because the AI invented fitment details.

Trust boundary: The prototype must not certify authenticity, guarantee vehicle compatibility, infer functionality from photographs, or present its checklist as legal or customs clearance. Unsupported claims must remain unverified, and listing approval must stay with the seller.

6. Shared Submission and Evaluation Framework

Participant deliverables

Each team should submit a **working prototype**, a **five-minute demonstration**, and a short supporting note covering the business problem, data used, approach, test results, limitations, and requirements for a real-world pilot. Source code and setup instructions should enable judges to reproduce the demonstration.

The demonstration should include one normal journey and one difficult case involving missing information, conflicting evidence, or an inappropriate requested action.

Judging criteria

Criterion	Weight	What judges should assess
Business relevance and value	30%	Does the solution address a meaningful seller problem with a measurable benefit?

AI quality and evidence	25%	Are outputs accurate, traceable, and appropriately cautious when information is missing?
Working prototype	20%	Does the end-to-end journey work, including difficult cases?
Seller experience	15%	Is the solution understandable, accessible, and practical for the intended user?
Responsible design and scalability	10%	Are privacy, human approval, operating cost, and future deployment considered?
Total	**100%**	

Common operating boundaries

Use synthetic data or explicitly approved, anonymised data; no production credentials or identifiable seller or buyer information should be required. Mock integrations are acceptable. Policy-dependent guidance must reference the supplied, dated source material, and unsupported conclusions must be labelled clearly.

Before sharing the briefs with participants, eBay and DCC should confirm the common datasets, reference checklists, mentor availability, and submission format. A shared test pack will make the competition more practical and judging more consistent.